

02 Cloud Computing Introduction

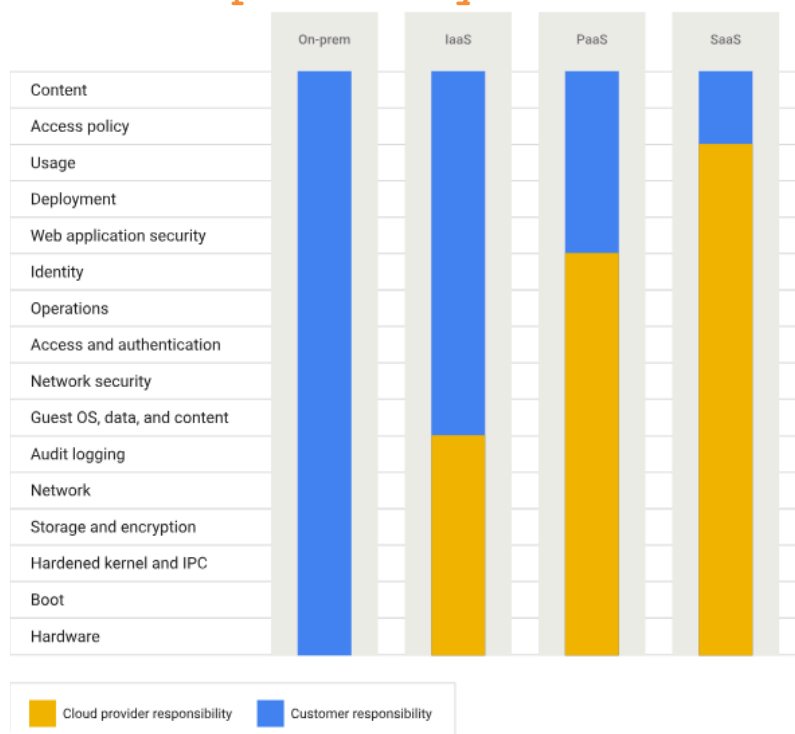
Cloud Computing Introduction

- Cloud Computing is the on-demand delivery of IT resources over the internet
- In essence, instead of buying physical infrastructure like servers, computers, managing a physical area to protect & manage this infrastructure – we simply rent & access these.
- Examples of IT Resources are Virtual Machines, Databases, Networking. The cloud services go one step beyond and include Internet of Things (IoT), Machine Learning (ML) and Artificial Intelligence (AI)

Benefits

- Using someone else's infrastructure. They have the headache of managing and maintaining it.
- The power to deploy databases, virtual machines, networks etc. at the click of a button!
- You lower your upfront costs, the infrastructure runs more efficiently, and can scale as per your need!

Shared Responsibility



Cloud Service Models and GCP Examples

Infrastructure as a Service (IaaS)

- The most flexible category that gives you maximum control over your cloud resources.
- The CSP is responsible for physical security, connectivity to internet, and the hardware only – you control everything else.
- Use Cases: Development and Testing Environments, Hosting Web Applications, Lift & Shift migrations

GCP IaaS Examples:

1. **Google Compute Engine:** Scalable, high-performance virtual machines.
2. **Google Cloud Storage:** Unified object storage.
3. **Google Cloud VPC:** Virtual Private Cloud to isolate resources.
4. **Google Persistent Disk:** Block storage for VMs.
5. **Google Cloud Load Balancing:** Scalable load balancing.
6. **Google Cloud Interconnect:** Dedicated connection to Google Cloud.
7. **Google Cloud DNS:** Scalable DNS service.

Platform as a Service (PaaS)

- A middle ground between IaaS and SaaS.
- The responsibility is split between you and the cloud provider.
- Allows you to focus more on development & go-live than patching & maintaining infrastructure.
- Think of it like a company PC – Hardware, OS, patches, databases maintained by IT support.
- Excellent for development frameworks and analytics!

GCP PaaS Examples:

1. **Google App Engine:** Platform for building scalable web applications.
2. **Google Cloud Functions:** Event-driven serverless compute service.
3. **Google Cloud SQL:** Managed relational database service.
4. **Google Cloud Run:** Managed compute platform for containers.
5. **Google Kubernetes Engine (GKE):** Managed Kubernetes service.
6. **Google Dataflow:** Stream and batch data processing.
7. **Google AI Platform:** Tools for building and deploying machine learning models.
8. **Vertex AI:** Platform for machine learning operations (MLOps).
9. **Google Maps Platform:** Geospatial services and APIs.

Software as a Service (SaaS)

- The most complete cloud service model, offering the highest level of managed services.
- While it provides the least flexibility in terms of underlying infrastructure, it often offers significant configurability within the application itself.
- It's the easiest to deploy, use, and go-live: almost like a ready-made app.
- Most of the responsibility is placed on the CSP; you are responsible for the data, applications & who has access.

GCP SaaS Examples:

1. **Google Workspace** (formerly G Suite):
 - Gmail for business
 - Google Drive
 - Google Docs, Sheets, and Slides
 - Google Meet for video conferencing
 - Google Calendar
2. **Google Cloud Identity:** Identity management service.
3. **Google Cloud Search:** Enterprise search for Google Workspace.
4. **Apigee API Management:** API management platform.
5. **Google Cloud Healthcare API:** Healthcare-specific data integration capabilities.

Deployment Models for Cloud Computing

1. Private Cloud

Definition: A private cloud is dedicated infrastructure operated solely for a single organization. It can be managed internally by the organization or by a third-party provider.

Characteristics:

- **Dedicated:** Resources are exclusively used by one organization, providing enhanced privacy and security.
- **Isolation:** Resources are dedicated to a single organization, ensuring better control over security and compliance.
- **Control:** Organizations have full control over the infrastructure, allowing customization and adherence to specific compliance requirements.
- **Cost:** Generally higher initial investment due to dedicated hardware and maintenance costs.

Examples:

- **Enterprise Private Cloud:** Large corporations may build their own data centers or use dedicated facilities to host their private cloud infrastructure. For example, financial institutions or government agencies that handle sensitive data often opt for private clouds.
- **Managed Private Cloud:** Organizations can also leverage managed service providers (MSPs) to operate and maintain their private cloud environments. For instance, VMware provides solutions for building and managing private clouds.

2. Public Cloud

Definition: A public cloud offers computing services over the internet to multiple users or organizations. Resources are shared across multiple clients, making it a cost-effective and scalable option.

Define Cloud Models

Public Cloud

The public cloud is a cloud computing model where cloud services are owned and provided by third-party vendors over the internet. These services are available to anyone who wants to use them, making it a versatile and cost-effective option for various purposes. With a public cloud, all hardware, software, and other supporting infrastructure are owned and managed by the cloud provider.

In a public cloud, you share the same hardware, storage, and network devices with other organizations or cloud "tenants," and you access services and manage your account using a web browser.

General public availability is a key difference between public and private clouds.

Key Characteristics of Public Cloud:

1. Shared Infrastructure:

Resources are shared among multiple users and organizations, optimizing utilization and cost.

2. Cost Efficiency:

Users only pay for the resources they consume, eliminating the need for upfront investments.

3. Scalability:

Public cloud services can easily scale up or down to accommodate changing demands.

4. Accessibility:

Services are accessible to the general public over the internet.

5. Examples:

Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), etc.

Characteristics:

- **Shared Infrastructure:** Resources (servers, storage, networking) are shared among multiple tenants.
- **Scalability:** Easily scale resources up or down based on demand, paying only for what you use.
- **Cost:** Lower upfront costs and operational expenses compared to private clouds.

Examples:

- **Amazon Web Services (AWS):** AWS provides a wide range of cloud services, including computing power, storage, and databases, accessible to the public.
- **Microsoft Azure:** Azure offers cloud services for computing, analytics, storage, and networking, used by businesses worldwide to build, deploy, and manage applications.
- **Google Cloud Platform (GCP):** GCP provides cloud computing services similar to AWS and Azure, with a focus on data analytics and machine learning.

Hybrid Cloud

A hybrid cloud is a combination of public and private clouds, designed to allow data and applications to be shared between them. This model offers flexibility and allows organizations to balance the benefits of both public and private environments.

For many organizations, a hybrid cloud approach is a MUST due to regulatory and data sovereignty requirements, tackling low latency issues etc.

Simply put – A private deployment can be used for the extra layer of security, and it can be coupled with a public deployment to handle surge in traffic / computing needs!

Key Characteristics of Hybrid Cloud:

1. Flexibility:

Organizations can choose where to host different workloads based on factors like security, performance, and cost.

2. Scalability:

Hybrid clouds can handle spikes in demand by utilizing resources from both environments.

3. Data Mobility:

Data can be moved seamlessly between private and public clouds as needed.

4. Cost Optimization:

Organizations can leverage cost-effective public cloud resources while maintaining sensitive workloads on-premises.

5. Example Use Case:

Healthcare organizations might use a private cloud for patient data storage while using a public cloud for non-sensitive applications.

Examples:

- **Financial Services:** Banks may use a private cloud for secure transactions and customer data while using public cloud services for customer-facing applications and analytics.
- **Healthcare:** Healthcare providers may store sensitive patient information on-premises or in a private cloud for compliance, while leveraging public cloud resources for research and collaboration.
- **Retail:** Retailers may use a hybrid cloud to manage their e-commerce platform on the public cloud while keeping sensitive inventory and customer data in a private cloud.

Define Cloud Models

Multi Cloud

- Slowly becoming more and more popular!
- In this kind of deployment, you utilize multiple Public Cloud Providers.
- Possible reasons:
 - Your organization doesn't solely want to rely on one CSP only.
 - Your organization started with one CSP but now wants to fully migrate to another CSP.
 - Your organization wants to utilize services/tools from different CSPs.

GCP

Stands out for its expertise in data management, analytics, and machine learning. They leverage Google's powerful internal infrastructure for high-performance and scalability.

AWS

Infrastructure as a Service (IaaS)

Definition: This is the most basic layer, providing virtualized computing resources like servers, storage, and networking. Users have control over the operating system and applications they install.

AWS IaaS Examples:

1. **Amazon EC2 (Elastic Compute Cloud):** Virtual servers in the cloud.
2. **Amazon EBS (Elastic Block Store):** Scalable block storage for use with EC2 instances.

3. **Amazon S3 (Simple Storage Service):** Scalable object storage.
4. **Amazon VPC (Virtual Private Cloud):** Isolated cloud resources.
5. **AWS IAM (Identity and Access Management):** Securely manage access to AWS services and resources.
6. **AWS Direct Connect:** Dedicated network connection to AWS.
7. **AWS Snowball:** Data transfer service using physical storage devices.

Azure IaaS Examples:

1. **Azure Virtual Machines:** On-demand scalable computing resources.
2. **Azure Blob Storage:** Massively scalable object storage.
3. **Azure Virtual Network:** Networking services to establish secure connections.
4. **Azure Disk Storage:** Managed disks for Azure VMs.
5. **Azure Load Balancer:** Distributes traffic among multiple servers.
6. **Azure Backup:** Backup solution for data protection.
7. **Azure Site Recovery:** Disaster recovery service.

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Platform as a Service (PaaS)

Definition: This layer provides a platform for developing, deploying, and managing applications. PaaS includes IaaS functionalities along with development tools and middleware.

AWS PaaS Examples:

1. **AWS Elastic Beanstalk:** Easy deployment and scaling of web applications.
2. **AWS Lambda:** Run code without provisioning servers.
3. **Amazon RDS (Relational Database Service):** Managed relational database.
4. **AWS Glue:** Managed ETL service.
5. **AWS Fargate:** Serverless compute engine for containers.
6. **AWS App Runner:** Fully managed service for web applications and APIs.
7. **Amazon SageMaker:** Build, train, and deploy machine learning models.

Azure PaaS Examples:

1. **Azure App Service:** Platform for building, deploying, and scaling web apps.
2. **Azure Functions:** Serverless compute service.
3. **Azure SQL Database:** Managed relational database as a service.
4. **Azure Logic Apps:** Automate workflows and business processes.
5. **Azure Kubernetes Service (AKS):** Managed Kubernetes container service.
6. **Azure Synapse Analytics:** Analytics service for data integration and big data.
7. **Azure Machine Learning:** Build and deploy machine learning models.

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Software as a Service (SaaS)

Definition: This layer delivers complete applications over the internet. Users access the software through a web browser or mobile app and don't have to worry about managing the underlying infrastructure.

AWS SaaS Examples:

1. **Amazon WorkSpaces:** Managed, secure Desktop-as-a-Service (DaaS).
2. **Amazon Chime:** Communications service for online meetings.
3. **Amazon QuickSight:** Business intelligence service.
4. **Amazon Connect:** Cloud-based contact center service.
5. **AWS Marketplace:** Online store for software and services.
6. **Amazon Polly:** Text-to-speech service.
7. **Amazon Comprehend:** Natural language processing service.

Azure SaaS Examples:

1. **Microsoft Office 365:** Cloud-based productivity suite.
2. **Microsoft Dynamics 365:** Cloud-based business applications.
3. **Azure DevOps:** Services for collaborative software development.
4. **Azure Cognitive Services:** Suite of AI services and APIs.
5. **Azure Security Center:** Unified security management.
6. **Power BI:** Business analytics service.
7. **Azure Sentinel:** Cloud-native security information and event management (SIEM).

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